

Functional Requirements

Functional requirements describe the desired end function of a system and any steps needed to reach that end state.

- **FR-01: Intrusion Simulation and Detection**
 - When a training session is active, the system shall generate a simulated intrusion event and classify it as one of three supported intrusion types.
- **FR-02: Alert Notification**
 - When a simulated intrusion event is generated, the system shall display an alert notification to the user.
- **FR-03: Training Profile Selection**
 - Before a session begins, the system shall require the user to select a training profile, and the system shall configure guidance and difficulty settings based on that selection.
- **FR-04: Randomized Attack Triggering**
 - During an active session, the system shall trigger intrusion events at random time intervals.
- **FR-05: Investigation Workflow**
 - When an alert is displayed, the system shall allow the user to select an investigation option and view relevant intrusion details.
- **FR-06: Response Selection**
 - After reviewing intrusion details, the system shall allow the user to select one response option from multiple available choices.
- **FR-07: Response Evaluation**
 - When a response is selected, the system shall evaluate the response and determine whether it is correct or incorrect.
- **FR-08: Escalation Behavior**
 - If an incorrect response is selected, the system shall increase the severity level of the intrusion event.
- **FR-09: Outcome Display**

- After evaluating a response, the system shall display the resulting state of the intrusion event.
- **FR-10: Explanation Feedback**
 - After displaying the outcome, the system shall provide an explanation indicating why the selected response succeeded or failed.
- **FR-11: Points Reward**
 - If a correct response is selected, the system shall award points to the user.
- **FR-12: Progress Tracking**
 - During a session, the system shall record user responses and update the user's progress metrics accordingly.
- **FR-13: Streak Tracking**
 - If consecutive correct responses occur, the system shall update and display the user's streak count.
- **FR-14: Alert Dismissal**
 - If the user selects a dismissal option, the system shall remove the alert from the active alert list and log the action.